MILITARY SPECIFICATION

SIGNAL, WARNING, AUDIBLE, FOR HEADSET, TYPE MA-1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of audible warning signal designated MA-1 for operation on 28 volts direct current (DC).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Setup
PPP-T-60	Tape: Pressure-Sensitive Adhesive; Waterproof, For Packaging

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specificaiton For
MIL-L-10547	Liners, Case And Sheets Overwrap; Water-Vaporproof Or Waterproof, Flexible

STANDARDS

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of

FSC 6340

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MIL-STD-704	Electric Power, Aircraft Characteristics And Utilization Of
MIL-STD-781	Reliability Test: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS3106	Connector, Plug, Electric, Straight, Solder Contacts, AN Type

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

3. REQUIREMENTS

3.1 Qualification. The warning signal furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Data. Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in Section 2 (see 6.2).

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

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* 3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.4.3 Materials shall conform to applicable specifications and shall be as specified herein. Materials that are not covered by specifications or are not specifically described herein shall be of the best quality, of the lightest practicable weight, and shall be suitable for the purpose intended.

3.5 Design and construction. The warning signal shall be hermetically sealed, compact, and sufficiently rugged to withstand shock and stresses incident to use on aircraft. Due to the high operating temperature requirement, the use of germanium diodes or transistors shall not be permitted. The design shall be completely composed of solid-state circuitry and shall not contain moving parts.

3.5.1 Adjustment. The warning signal shall be permanently adjusted at the time of manufacture and shall not require further adjustments.

3.5.2 Connections. Connections to the warning signal shall be made as shown on Figure 1.

3.5.2.1 Safety wiring. Means shall be provided for body attachment of safety wiring.

3.5.3 Input transient voltage protection. Components of the warning signal shall be protected against high-voltage spikes of 200 volts DC of 1 microsecond time duration, and against transients, as specified in MIL-STD-704.

3.6 Performance.

3.6.1 Frequency and interruption rate of the warning signal. The frequency and interruption rates shall be:

a. Fundamental audio output frequency 250 ±50 Hertz (Hz).

b. Interruption rate 5 ±1 Hz.

* 3.6.1.1 Output audio cycle. When interrupted at the specified rate, the On-Off cycle of the audio output shall be 50 percent On, 50 percent Off, with a 5 percent tolerance in the On/Off cycle.

3.6.1.2 Harmonic content. The warning signal, when energized with 28 volts DC and loaded as specified in 3.6.2.2, shall have an output signal containing both odd and even harmonics. The voltage readings of harmonics 2 through 11 shall total 120 to 220 percent of the fundamental voltage reading. The voltage readings of the odd harmonics shall total a minimum of 75 percent of the total harmonic count, excluding the fundamental.

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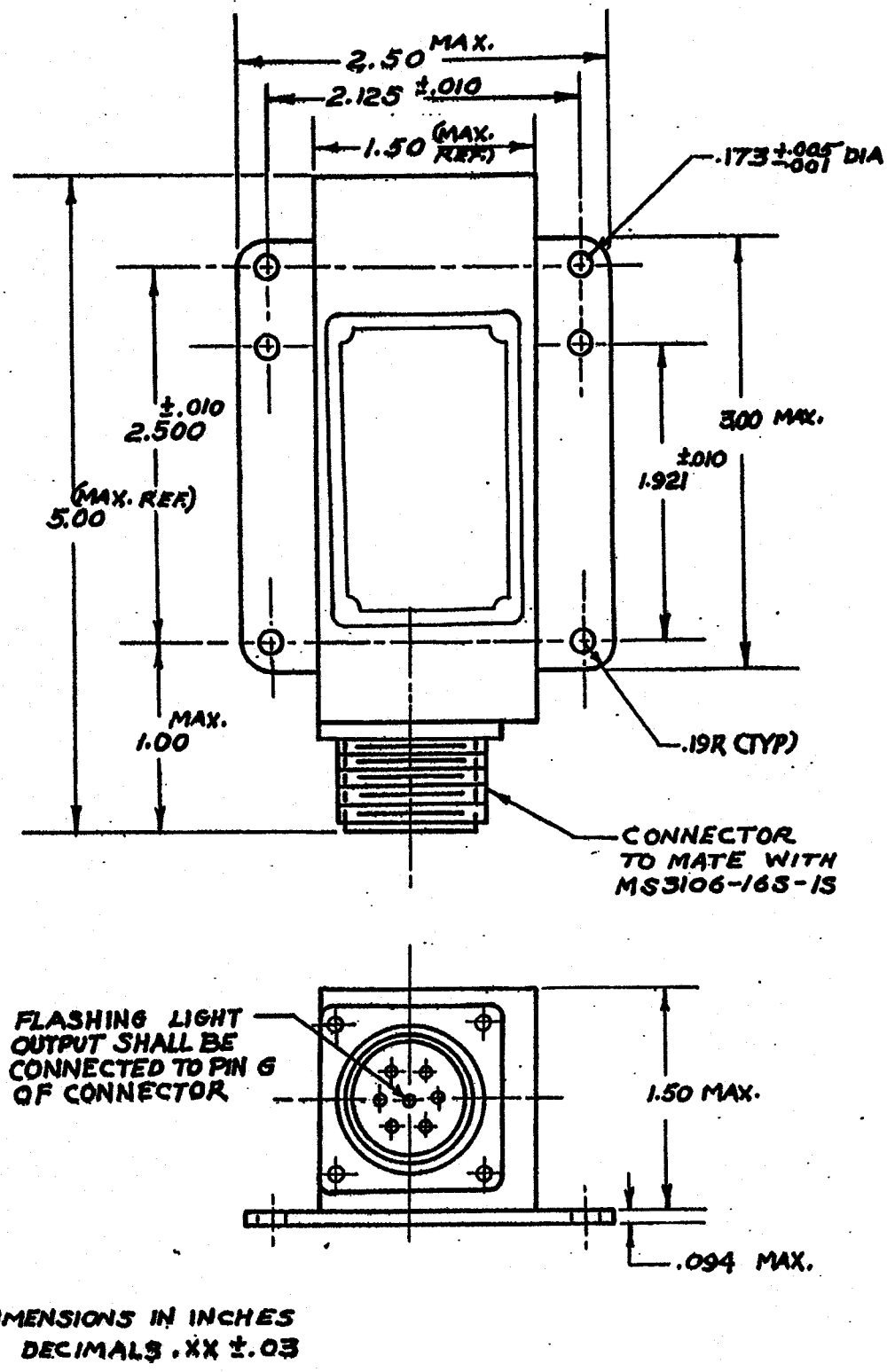


FIGURE 1. Warning Signal MA-1 Envelope

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* 3.6.13 Audio and flashing lamp output circuits. The interrupted audio output and flashing lamp output shall be shown below. The power measured shall be true root-mean square power. The 0 db shall be the base of 1 milliwatt into 150 ohms. The flashing rate of the lamp (MS-type) shall be 5 ±1 Hz.

Connector Pin		DB Level	Flashing Rate
C- and ground	B	22 ±2	
D- and ground		16 ±3	
E- and ground		9 ±3	
G- and ground		Flashing lamp output	5 ±1 Hz

* 3.6.1.4 Steady tone. A steady tone of the specified frequency, uninterrupted, shall be provided between connector terminals F and ground when the signal is energized. The output of this tone shall be 19 ±3 decibel (db). The 0 db shall be the base of 1 milliwatt.

* 3.6.1.5 Flashing lamp circuit. When interrupted at the specified rate, the On-Off cycle of the flashing light shall be 50 percent On, 50 percent Off, with a 5 percent tolerance in the On/Off cycle.

3.6.2 Electrical circuits.

3.6.2.1 Power input. Terminal A of the connector shall be connected to 28 volts DC positive. The unit shall be energized by connection of terminal B through external circuitry to 28 volts DC negative.

* 3.6.2.2. System loading. The warning signal shall be compatible with the aircraft communication system. The portion of the communication system into which the warning signal shall operate is essentially an audio transformer terminated in a 9 ohms resistive headset. The transformer should have the following characteristics: Line type, primary impedance 150 ohms, primary DC resistance 7.5 ohms maximum, secondary impedance 9 ohms, secondary DC resistance 4.5 ohms maximum, a frequency response of 200-6000 Hz, and an output of 200 ±5 milliwatts.

3.6.2.3 Electrical output characteristics. When not energized, the warning signal shall present an open circuit to associated communications equipment. The resistance between any output circuit and terminal A of the connector shall be not less than 10 megohms. When the signal warning is energized, it shall be impossible for uninterrupted direct current to be conducted into the associated communication equipment.

3.6.3 Operating voltage. The performance specified in 3.6.1 shall be met over an input voltage range from 27 to 29 volts DC. Over the input voltage range between 14 to 29 volts DC, the warning signal shall operate without damage to its component.

3.6.4 Current consumption. When operating on 29 volts DC, the average current consumed by the warning signal shall not exceed 0.20 amperes.

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* 3.6.5 Environmental conditions. The warning signal shall operate electrically without failure under the following conditions:

a. Temperatures: -65° to +125° Centigrade (C).

b. Vibration: Typical aircraft vibration. Vibration isolators shall not be used. (See 4.6.9)

c. High altitude - low temperature: Pressure equivalent to 80,000 feet altitude and temperature of -65°C.

d. Shock: 50 Gravity.

3.6.6 Dielectric strength. The warning signal shall withstand a voltage of 600 volts root mean square for 60 seconds when applied simultaneously between all connector terminals and the case.

3.6.7 Hermetic seal. The warning signal shall be hermetically sealed to withstand a pressure of 20 millimeter of mercury while submerged in water or ethylene glycol.

* 3.6.8 Short circuit. The warning signal and audio transformer shall withstand a short circuit for a minimum of 30 seconds from any output pin to ground.

3.6.9 Pressurization. The warning signal shall withstand pressure of 28 pounds per square inch.

* 3.6.10 Reliability. Unit shall exhibit a minimum acceptable mean time between failure of 500 hours at 90 percent confidence level.

* 3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.8 Dimensions. The warning signal shall conform to the dimensions shown on Figure 1.

3.9 Weight. The weight of the warning signal shall be not more than 6 ounces.

3.10 Screw threads. Machine screw threads shall conform to MIL-S-7742.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship. The warning signal, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, accuracy of dimensions, and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all the inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities

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suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the signal shall be classified as follows:

a. Qualification tests (4.4)

b. Quality conformance tests (4.5)

4.3 Test conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure of approximately 29.92 inches mercury and a temperature of 25° ±5°C.

4.4 Qualification testing.

4.4.1 Test samples. The qualification test samples shall consist of five signals representative of the production equipment. The samples shall be identified with the manufacturer's part number and any additional information required by the procuring activity.

4.4.2 Test report. After the contractor completes the qualification tests, he shall prepare a test report in accordance with MIL-STD-831 and furnish three complete copies to the procuring activity.

4.4.3 Qualification tests. Qualification tests shall consist of all the tests specified under 4.6. At the conclusion of each environmental test, the signal shall be tested for operation and audio output. The tests shall be performed on the samples as indicated and in the order listed in Table I.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests (4.5.1)

b. Sampling tests (4.5.2)

4.5.1 Individual tests. Each signal shall be subjected to the following tests as described under 4.6:

a. Examination of product (4.6.1)

b. Dielectric strength (4.6.2)

c. Current consumption (4.6.3)

d. Operation (4.6.4)

* 4.5.2 Sampling tests. Sampling shall be in accordance with MIL-STD-105, special inspection level S-4, AQL 2.5 percent.

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TABLE I. Qualification Test Schedule

Test	Sample Number				
	1	2	3	4	5
a. Individual tests	X	X	X	X	X
b. Internal impedance	X	X	X	X	X
c. Audio outputs circuits	X	X	X	X	X
d. Frequency and interruption rate	X	X	X	X	X
e. Reliability	X			X	
f. Vibration		X			
g. Shock					X
h. High altitude-low temperature			X		
i. Pressurization			X	X	
j. High temperature					X
k. Low temperature		X			
l. Short circuit	X	X			
m. Harmonic content	X	X	X	X	X
n. Hermetic seal			X	X	

4.5.2.1 Lot. A lot shall consist of warning signals manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

* 4.5.2.2 Tests. Unless ohterwise specified by the procuring activity, the number of warning signals specified shall be selected at random from the specified lot size and subjected to:

a. Individual tests (4.5.1)

b. Internal impedance (4.6.5)

c. High temperature (4.6.13)

d. Reliability. Test per Test Plan XXVII, fixed length production verification test, MIL-STD-781 (4.6.8)

e. Vibration (limit test to room temperature) (4.6.9)

f. Shock (4.6.10)

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4.6 Test methods.

4.6.1 Examination of product. The warning signal shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, dimensions, weight, and marking.

4.6.2 Dielectric strength. The warning signal shall be subjected to a voltage of 600 volts root mean square for 60 seconds. The voltage shall be applied simultaneously between all terminals of the connector and the case. At the conclusion of this test, there shall be no evidence of breakdown and the warning signal shall operate satisfactorily.

4.6.3 Current consumption. The output pins of the warning signal shall be connected to a 150-ohm load and the current measured to insure that the current does not exceed 0.20 ampere when operated at 29 volts DC. This test shall be performed separately on each output pin.

4.6.4 Operation. The warning signal shall be monitored with an oscilloscope to determine its operation from a 14 to 29 volts DC source. Frequency, db level, and interruption rate shall be monitored when the warning signal is loaded as specified in 3.6.2.2.

4.6.5 Internal impedance. The impedance between pin A and each output pin shall be measured with pin A connected to 28 volts DC positive, pin B disconnected, and an ammeter connected between the output terminal and ground. A DC microammeter shall be used and the maximum current shall not exceed 2.8 microamperes.

* 4.6.6 Audio and flashing lamp output circuits. The db levels indicated in 3.6.1.3 and 3.6.1.4 shall apply. The levels shall be measured with a thermovoltmeter connected across a 150-ohm resistive load. The 150-ohm load shall be the combined parallel impedance of the thermovoltmeter used and the load resistor. The voltage readings shall be converted to db readings. The 0 db shall be the base of 1 milliwatt into 150 ohms load. Direct current of 28 volts shall be applied to the warning signal. The On-Off cycle indicated in 3.6.1.5 shall apply for the flashing light output circuit. Terminal G of the connector shall be connected to the flasher circuitry internally and to an MS type lamp externally.

4.6.7 Frequency and interruption rate. When loaded as specified in 3.6.2.2, the requirement of 3.6.1 and 3.6.1.1 shall be met and tested by monitoring the output warning signal with an oscilloscope.

* 4.6.8 Reliability. Unit shall be tested per Test Plan I, MIL-STD-781, with a duty cycle of 250 hours continuous operation, followed by 250 hours cycling, 1 minute on and 1 minute off, at a normal input voltage and simulated load of 150 ohms bsetween pin C and ground. Failure to meet requirements regarding output shall be considered a failure.

* 4.6.9 Vibration. The warning signal shall be subjected to a vibration test in accordance with Method 514, Category (a) of MIL-STD-810.

* 4.6.10 Shock. The warning signal shall be subjected to a shock test in accordance with Method 516 of MIL-STD-810, except that the magnitude of each shock impulse shall be 50 Gravity.

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4.6.11 High altitude -low temperature. The warning signal shall be placed in in an altitude chamber maintained at a temperature of -65°C and the pressure reduced to simulate an altitude condition of 80,000 feet for 4 hours. After returning to room temperature, the warning signal shall be examined for any evidence of damage and shall operate as specified herein.

4.6.12 Hermetic seal. The warning signal shall be placed in a transparent container filled with ethylene glycol or water and placed in an altitude chamber equipped with a view plate or bell jar. The pressure shall be reduced to 20 millimeter mercury or less (approximately 81,000 feet altitude) and maintained at this pressure for 30 minutes. Any leakage as indicated by bubbles shall constitute failure.

* 4.6.13 High temperature. The warning signal shall be subjected to a low temperature test in accordance with Method 501 of MIL-STD-810, except that the temperature shall be 125°C. During the final hour of this test, the signal shall operate as specified herein.

* 4.6.14 Low temperature. The warning signal shall be subjected to a low temperature test in accordance with Method 502 of MIL-STD-810, except that the temperature for the entire test shall be -54°C. During the final hour of the test, the warning signal shall be energized and shall operate as specified herein.

4.6.15 Short circuit. All output pins of the warning signal shall be individually connected to ground and the warning signal energized for not less than 30 seconds. Upon completion of this test, the warning signal shall operate as specified herein.

4.6.16 Harmonic content. The warning signal shall be tested to assure compliance with the requirements of 3.6.1.2. The above measurements shall be made on all four output pins separately, using a harmonic analyzer connected across the secondary of the audio transformer when loaded as specified in 3.6.2.2.

4.6.17 Pressurizaiton test. The warning signal shall be placed in a chamber with absolute pressure of 28 pounds per square inch and a temperature of 25°C for 4 hours. During this period, no deformation or other damage to the warning signal shall occur. At the conclusion of this test, the warning signal shall be closely examined for evidence of deformation and shall operate as specified herein. Evidence of deformation, damage or any unsatisfactory operation shall be cause for rejection.

* 4.6.18 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

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* 5.1.1 Level A.

* 5.1.1.1 Cleaning. The warning signals shall be cleaned in accordance with process C-1 of MIL-P-116.

* 5.1.1.2 Drying. The warning signals shall be dried in accordance with one or more of the drying procedures listed in MIL-P-116.

* 5.1.1.3 Unit protection. The warning signals shall be preserved in accordance with Method III of MIL-P-116 and individually packaged in containers conforming to PPP-B-676. Sufficient cushioning shall be provided to prevent shock damage and prevent movement of the item within the package. Prior to packaging, the open end of the connector shall be sealed with tape conforming to PPP-T-60.

* 5.1.2 Level C. The warning signals shall be preserved and packaged to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice when it meets the requirements of this level.

* 5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. The warning signals, preserved and packaged in accordance with 5.1.1, shall be packed in shipping containers conforming to PPP-B-636, class weather resistant. When the weight limitation of PPP-B-636 is exceeded, the signals shall be packed in shipping containers conforming to PPP-B-601, overseas type, style optional. So far as practicable, exterior containers shall be of uniform shape and size and of minimum tare and cube consistent with the protection required, and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When a PPP-B-601 shipping container is used, a case liner conforming to MIL-L-10547 shall be provided.

* 5.2.2 Level B. Level B packing shall be the same as level A except the shipping container shall conform to PPP-B-636, class domestic or PPP-B-601, domestic type, style optional.

* 5.2.3 Level C. The warning signals shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

* 5.3 Physical protection. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186.

* 5.4 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. The nomenclature shall be as follows:

Signal, Warning, Audible, For Headset, Type MA-1 (With Visual Indicator Capabilities).

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6. NOTES

* 6.1 Intended use. The MA-1 warning signal is for use on aircraft to warn the pilot of an unsafe landing gear condition through injection of an interrupted tone in his headset and a flashing lamp indicator in the cockpit.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.

b. Data requirements (see 3.2).

c. Selection of applicable levels of preservation, packaging, and packing (see Section 5).

* 6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: MTE, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

* 6.4 International interest. Certain provisions (see 3.6.1, 3.6.1.1 and 3.6.1.2) of this specification are the subject of international standardization agreement ABC Air Standard 10/30D (STANAG 3370). When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activities:
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Air Force - 11, 82

International Interest (See 6.4)

Preparing Activity:
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